



Banking Customer Data Analysis Project

Exploratory Data Analysis & Business Intelligence Report

Dataset: Banking.csv — 3,000 customers | 25 features

Tools: Python (Pandas, Seaborn, Matplotlib) | Microsoft Power BI

Deliverables: Jupyter Notebook | Power BI Dashboard

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DATA ANALYTICS PORTFOLIO PROJECT

Business Problem Statement

Banks manage thousands of diverse customers with varying financial behaviors, income levels, and risk profiles. Without structured understanding of these segments, strategic decisions around product offerings, risk management, and retention become difficult.

Core Business Challenges

- Identifying high-value customers who drive disproportionate revenue
- Detecting customers at risk of churn or financial distress
- Optimizing loans, credit cards, and business lending
- Tailoring fee structures and loyalty programs to retain profitable segments
- Assessing portfolio risk across demographics and geographies

Research Questions

- What is the financial profile of the bank's customer base in terms of income, savings, loans, and credit usage?
- How do loyalty tier and fee structure vary across customer demographics (nationality, occupation, age)?
- What correlations exist between financial variables like income, deposits, loans, and business lending?
- Which customer segments carry the highest risk weighting, and what does that mean for the bank's risk exposure?
- How can the bank better segment customers to personalize financial products and improve loyalty outcomes?

Project Overview

This project combines rigorous exploratory data analysis with an interactive business intelligence dashboard to deliver actionable insights for banking stakeholders.

Domain

Banking / Financial Services

Project Type

Exploratory Data Analysis (EDA) +
Business Intelligence Dashboard

Dataset

3,000 customers, 25 features covering
demographics, financial holdings, and
loyalty

Analysis Tools

Python – Pandas, NumPy, Seaborn,
Matplotlib

Dashboard Tool

Microsoft Power BI

Deliverables

Jupyter Notebook (EDA) + Power BI
Dashboard (.pbix)

Status

Completed

Dataset Description

The dataset contains 3,000 banking customer records with 25 features covering demographics, financial holdings, product usage, and loyalty classification.

Feature Categories

Category	Columns
Identity	Client ID, Name
Demographics	Age, Nationality, Location ID, GenderId
Occupation & Contact	Occupation, Banking Contact, BRId, IAId
Financial Products	Credit Cards, Credit Card Balance, Bank Loans, Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Business Lending, Superannuation Savings
Classification	Fee Structure, Loyalty Classification, Risk Weighting, Properties Owned
Income	Estimated Income
Timeline	Joined Bank

Key Statistics

3,000 Customers 25 features per record	Age: 17–85 yrs Mean: ~51 years
Income: ₹15K–₹5.2L Mean: ₹1,71,305	Avg Business Lending ~₹8,66,760

Categorical Distributions

- **Loyalty:** Jade (44.4%) > Silver (25.6%) > Gold (19.5%) > Platinum (10.6%)
- **Fee Structure:** High (49.2%) > Mid (32.1%) > Low (18.7%)
- **Nationality:** European (43.6%) > Asian (25.1%) > American (16.9%) > Australian (8.5%) > African (5.9%)
- **Risk Weighting:** Level 2 (40.7%) most common; Level 5 (5.3%) least common
- **Properties Owned:** Roughly uniform across 0, 1, 2, and 3 properties

Credit Card Distribution

- 1 Card: 64%
- 2 Cards: 25.6%
- 3 Cards: 10.4%

Exploratory Data Analysis (EDA)

The full EDA is documented in `final_banking_EDA.ipynb` and covers the following analytical stages.

01

Data Loading & Initial Inspection

Loaded the dataset using Pandas and inspected shape: (3000, 25). Verified data types — a mix of object, int64, and float64 columns. Reviewed descriptive statistics using `.info()` and `.describe()`. Confirmed no structural issues; ID columns treated as categorical identifiers.

02

Feature Engineering

Created a new **Income Band** feature by binning Estimated Income into Low (< ₹1,00,000), Medium (₹1,00,000 – ₹3,00,000), and High (> ₹3,00,000). This derived variable enabled richer segmentation across loyalty, occupation, and nationality dimensions.

03

Univariate Analysis

Plotted count distributions for all categorical features using Seaborn countplot. Generated KDE-overlaid histograms for 8 numerical features: Age, Estimated Income, Superannuation Savings, Credit Card Balance, Bank Loans, Bank Deposits, Saving Accounts, and Business Lending.

04

Bivariate Analysis

Used Nationality as a hue variable across all categorical features to reveal cross-demographic behavioral patterns. Compared financial product usage across Loyalty Classification and Fee Structure tiers.

05

Correlation Analysis

Generated a correlation heatmap for 8 core financial variables using Seaborn heatmap with the crest color palette. Identified strong, moderate, and negligible correlations guiding product strategy recommendations.

Key Insights & Findings

Correlation Analysis Results

Variable Pair	r Value	Strength	Business Insight
Bank Deposits ↔ Saving Accounts	0.75	Strong	Deposit-heavy customers also maintain large savings accounts
Business Lending ↔ Bank Deposits	0.44	Moderate	Business borrowers tend to also deposit more
Business Lending ↔ Bank Loans	0.42	Moderate	Heavy product users cluster together
Credit Card Balance ↔ Bank Deposits	0.38	Moderate	Higher card users also deposit more
Estimated Income ↔ Superannuation Savings	0.37	Moderate	Higher earners save more for retirement
Estimated Income ↔ Bank Loans	0.33	Moderate	Higher income customers take more loans
Estimated Income ↔ Business Lending	0.33	Moderate	Wealthier customers use business credit more
Age ↔ All Financial Variables	~0.00	Negligible	Age is not a predictor of financial activity

 **Key Takeaway:** Age is not a meaningful segmentation variable for this bank's customers. Financial behavior is much better explained by income level, deposit volume, and degree of product engagement.

Loyalty & Fee Structure Observations

- Jade loyalty tier dominates at 44.4% — a significant upsell opportunity to Silver, Gold, or Platinum tiers
- High fee structure customers form the largest segment at 49.2%, indicating a premium-priced institution
- No single nationality is uniformly over-represented in premium loyalty tiers — income and product usage are stronger loyalty drivers than ethnicity

Product Usage & Risk Profile

- 64% of customers hold only 1 credit card — clear headroom for cross-selling additional credit products
- Business Lending varies widely (₹0 – ₹38.26 lakh), pointing to very different customer business engagement levels
- Foreign Currency Accounts are a relatively rare product — a niche upsell opportunity for high-income or internationally-oriented customers
- Risk Level 2 is most common (40.7%), indicating a moderately low-risk portfolio overall
- Fewer than 6% of customers sit at Risk Level 5 (highest risk), limiting exposure to high-risk defaults
- Risk weighting can potentially be cross-referenced with income bands for smarter loan approval workflows

Power BI Dashboard

The Power BI dashboard (`final_banking_dashboard.pbix`) delivers an interactive, visual summary of the analysis with the following components.



Customer Demographics

Age distribution, nationality breakdown, and gender split across the full customer base.



Financial Portfolio Summary

Income bands, deposits, loan values, and savings comparisons in a single view.



Loyalty & Fee Structure

Loyalty tier distribution by nationality and occupation with fee structure overlays.



Risk Profiling

Risk weighting distribution and correlation with financial holdings.



Product Usage Analysis

Credit card counts, foreign currency accounts, and business lending trends.



Interactive Filters & Slicers

Dynamic filtering by nationality, occupation, loyalty tier, and fee structure.

 To view the dashboard: Open `final_banking_dashboard.pbix` in Microsoft Power BI Desktop (free download available at powerbi.microsoft.com).

Conclusions & Recommendations

Key Conclusions

- The customer base is financially diverse but skews toward moderate-to-high income earners with consistent deposit behavior
- Income and product engagement – not age – are the primary drivers of financial activity
- The strong deposit-savings correlation (0.75) suggests a loyal, savings-oriented core segment that may respond well to wealth management products
- The majority of customers in the Jade loyalty tier represent a large addressable segment for loyalty program upgrades
- The bank's risk portfolio is generally healthy, with most customers at Risk Level 1–2

Strategic Recommendations

→ Cross-sell Business Lending

Strong correlation (0.44) between deposits and business lending usage

→ Upgrade Jade-tier Customers

44% sit at the lowest loyalty tier – large addressable upsell potential

→ Offer Retirement Products

Moderate income-superannuation correlation of 0.37 supports targeted outreach

→ Credit Card Bundle Campaigns

64% of customers hold only 1 credit card – clear cross-sell opportunity

→ Risk-Adjusted Product Tiers

Align product offerings with Risk Weighting scores (1–5)

→ Nationality-Targeted Campaigns

European (43.6%) and Asian (25.1%) are the two dominant segments

Future Scope

Churn Prediction

Build predictive models for customer churn using loyalty tier transitions

Credit Risk Scoring

Develop a credit risk scoring model using Risk Weighting, Income Band, and Loan data

Customer Clustering

Apply K-Means / Hierarchical clustering to derive data-driven customer segments

Lifetime Value Analysis

Integrate time-series analysis of the 'Joined Bank' field to study customer lifetime value trends